

Exploratory Data Analysis on the Automobiles Dataset

Report

Introduction

This dataset is a comprehensive catalogue of 205 automobiles, each described by 26 technical, performance, and pricing attributes. It mixes specifications, engine details, fuel economy, and market price. Each row represents a single car model configuration.

Data cleaning

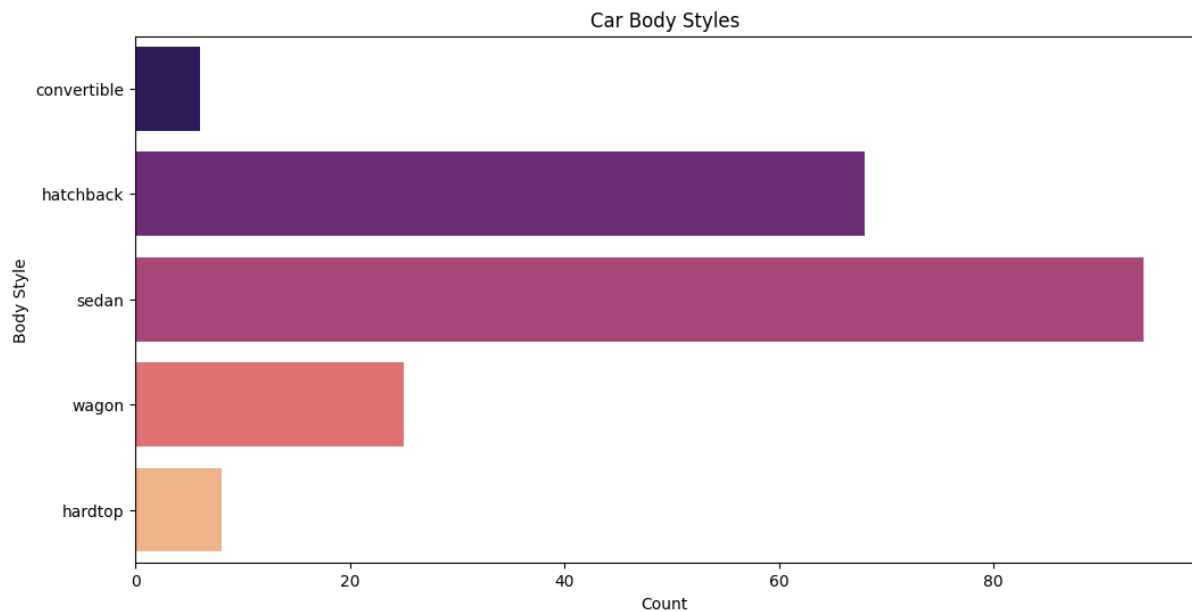
Since every vehicle in the dataset had its engine located at the front, the 'engine location' column offered no variability and was dropped to avoid carrying a constant, non-informative feature. I also removed 'normalised losses' and 'symbolizing' because they were not relevant to the analytical goals and would not contribute meaningful insight. To improve consistency and enable numerical operations, I converted the 'num of doors' field from categorical text labels into numeric values, ensuring it can be used effectively in visualisations and statistical summaries. Additionally, I identified and removed duplicate rows to prevent repeated observations.

Missing data

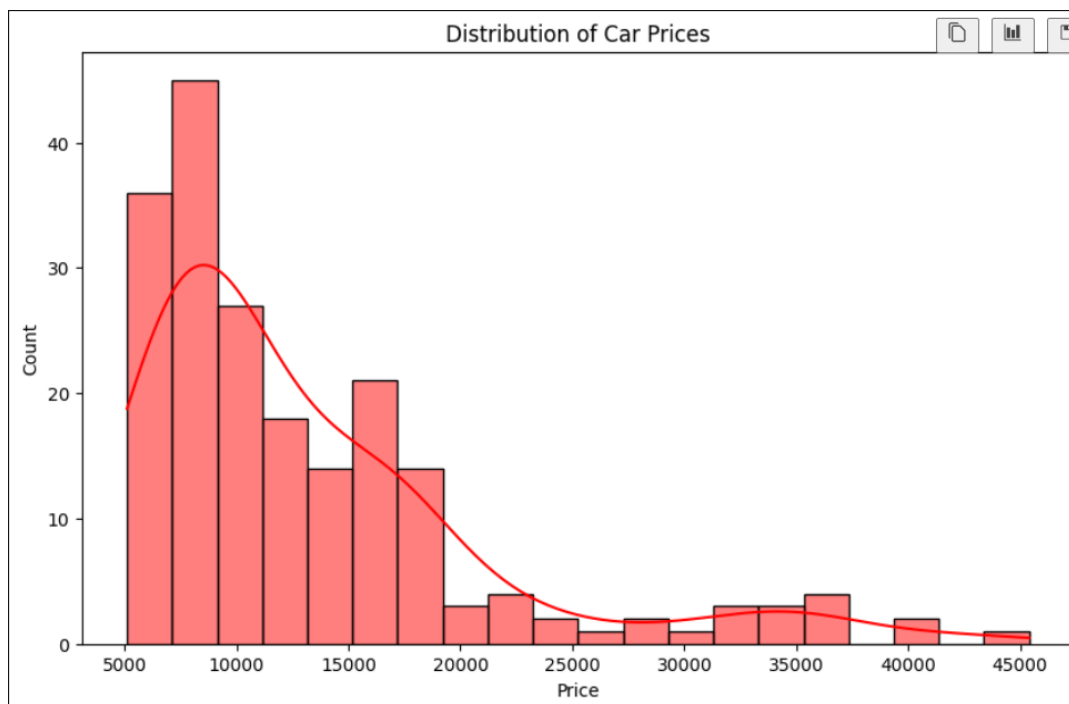
To handle the missing data, I converted all missing values recorded as "?" into NaN values to ensure they were recognised by Python's analytical tools. A summary of missingness revealed gaps in 'num of doors', 'bore', 'stroke', 'horsepower', 'peak rpm', and 'price'. For the numerical engine-related attributes (bore, stroke, horsepower, and peak rpm) I replaced missing entries with their respective mean values to maintain the overall distribution without introducing bias from extreme values. Since most vehicles in the dataset had four doors, I imputed missing values in 'num of doors' using the mode. For 'price', I removed all rows with missing values because price is the key target variable for comparison in the exploration. I converted all string data to numeric.

Data stories and visualisations

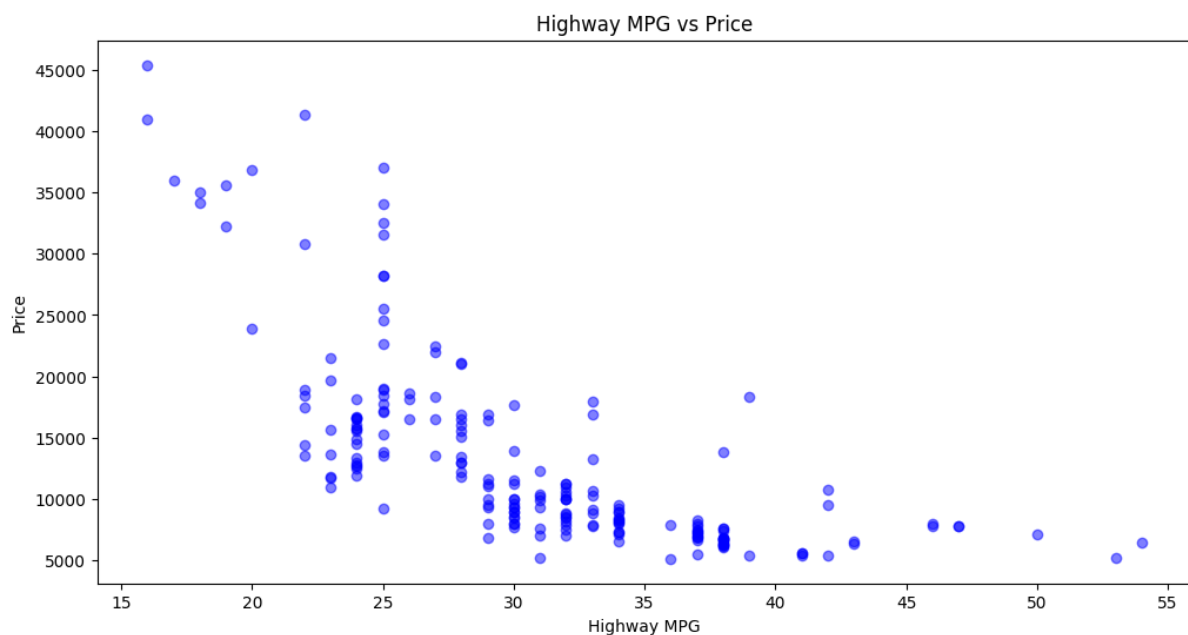
To begin exploring the dataset, I examined the distribution of vehicle body styles to understand which types dominate the sample. Visualising the frequency of each category made the pattern immediately clear: Sedans are the most common body style.



To explore price extremes in the dataset, I began by visualising the overall distribution of vehicle prices using a histogram. The histogram shows that car prices in the dataset are heavily concentrated at the lower end of the price range. Most vehicles fall between roughly £5,000 and £15,000, with the count dropping sharply as prices increase. The dataset is dominated by affordable, lower-priced cars, while high-priced vehicles are relatively rare.

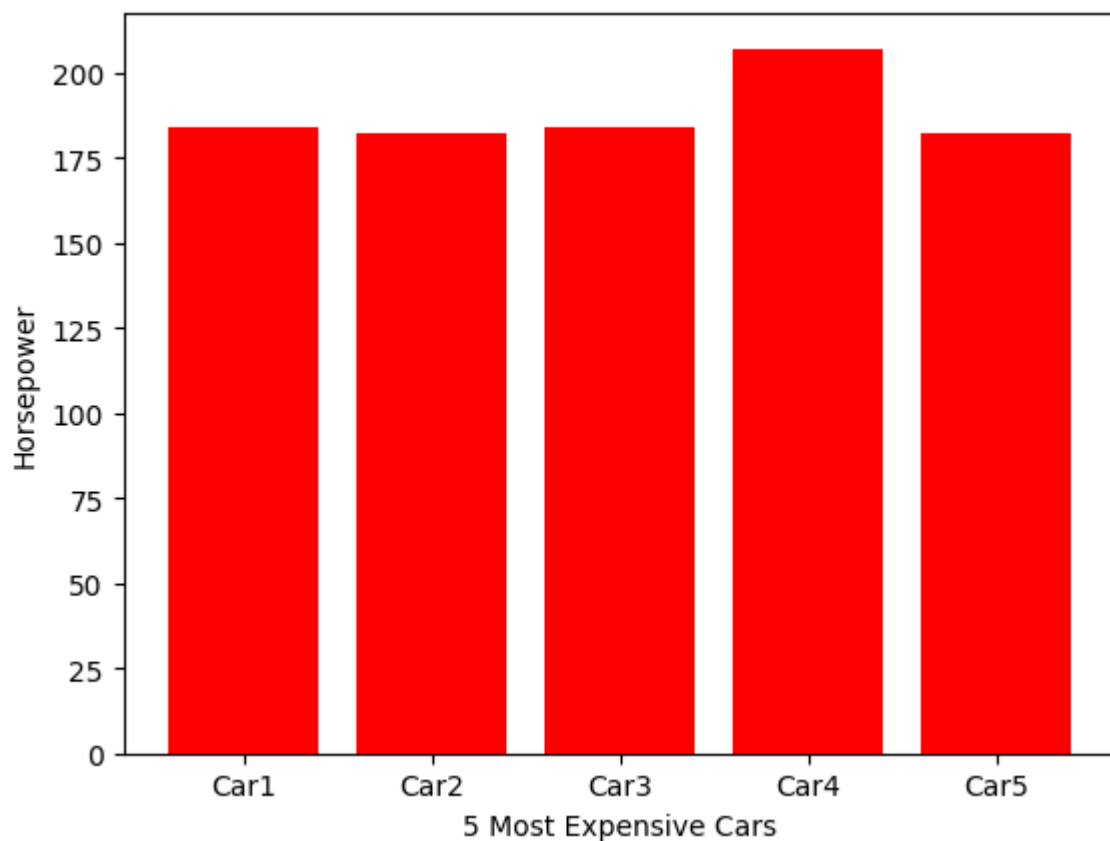


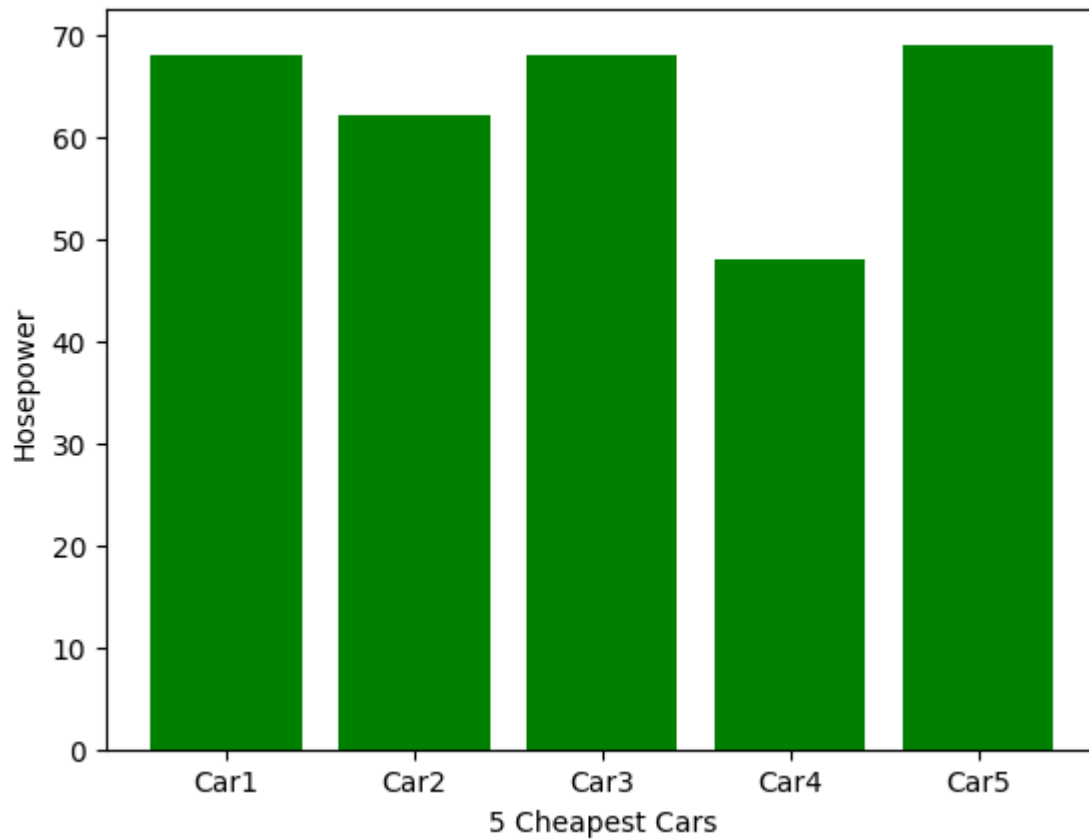
To understand how fuel efficiency relates to vehicle pricing, I plotted city MPG vs. price and highway MPG vs. price across all cars in the dataset. Both scatter plots revealed a consistent inverse relationship: lower-priced vehicles tend to achieve higher MPG, whether measured in city or highway conditions. This pattern aligns with expectations, as many budget-friendly models prioritise smaller engines and lighter builds, which naturally improve fuel economy. Conversely, higher-priced vehicles which are often larger, more powerful, or performance-oriented, cluster toward lower MPG values. This highlights a strong link between price and fuel efficiency.



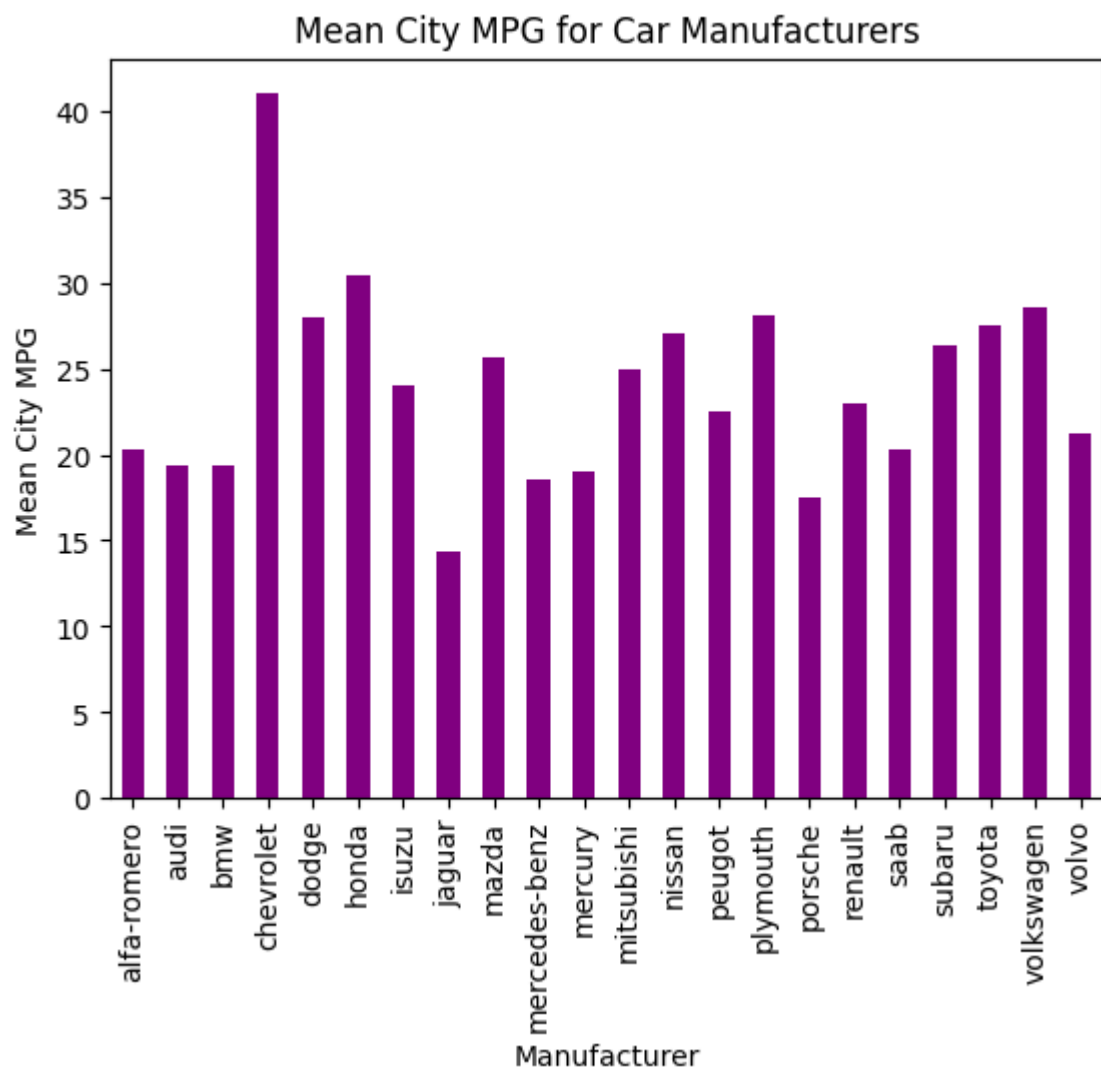
To deepen the analysis of how price relates to vehicle characteristics, I sorted the dataset from highest to lowest price and isolated the five most expensive and five least expensive cars for closer comparison. Examining their horsepower revealed that the

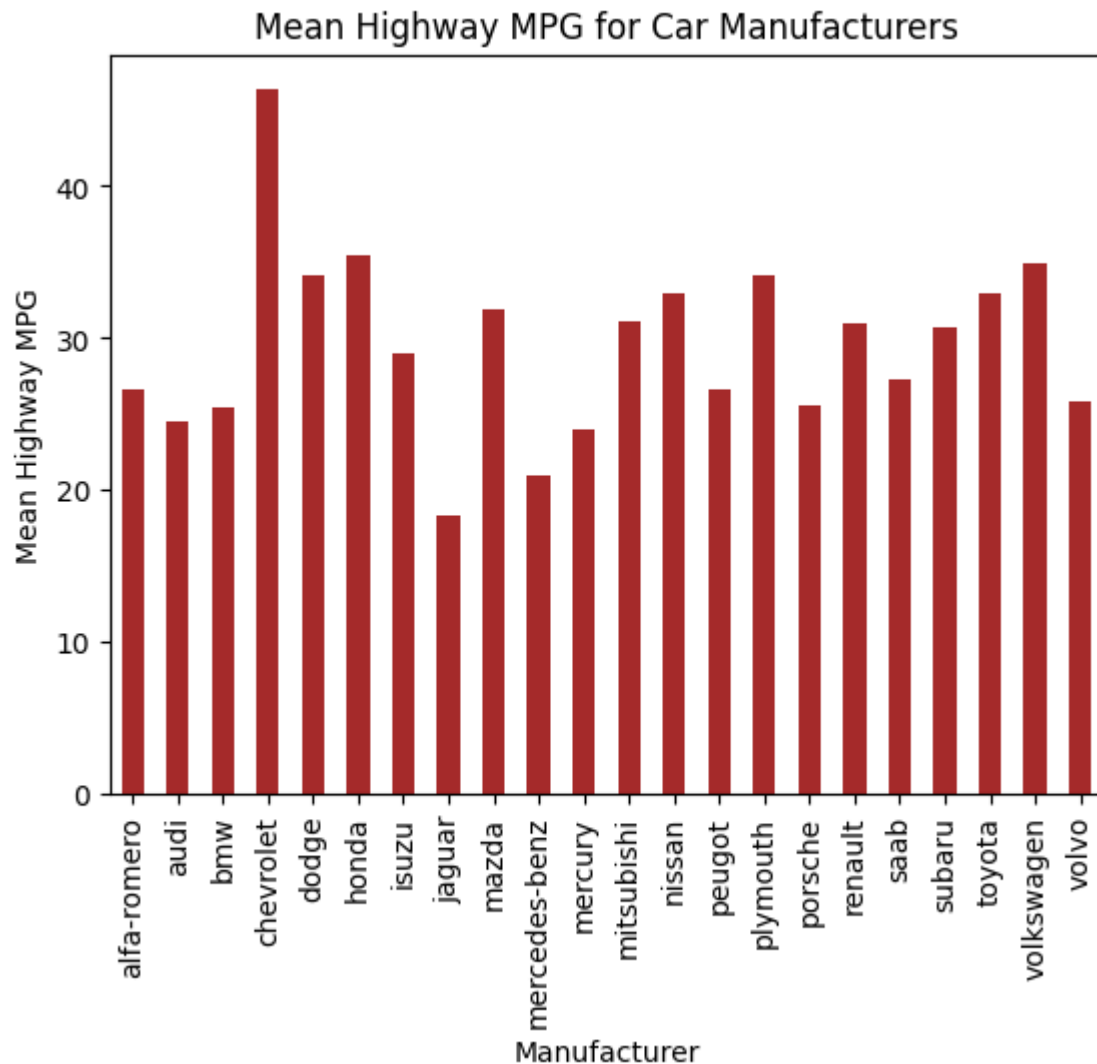
most expensive vehicles consistently fell within a relatively narrow band of roughly 175–250 horsepower, suggesting a shared emphasis on stronger performance. In contrast, the cheapest vehicles displayed both much lower horsepower values (around 45–70) and greater variation, indicating a mix of very small, efficiency-focused engines. This contrast reinforces the broader trend that higher prices in this dataset are closely tied to higher-powered engines, while the lowest-priced models prioritise affordability and fuel economy over performance.



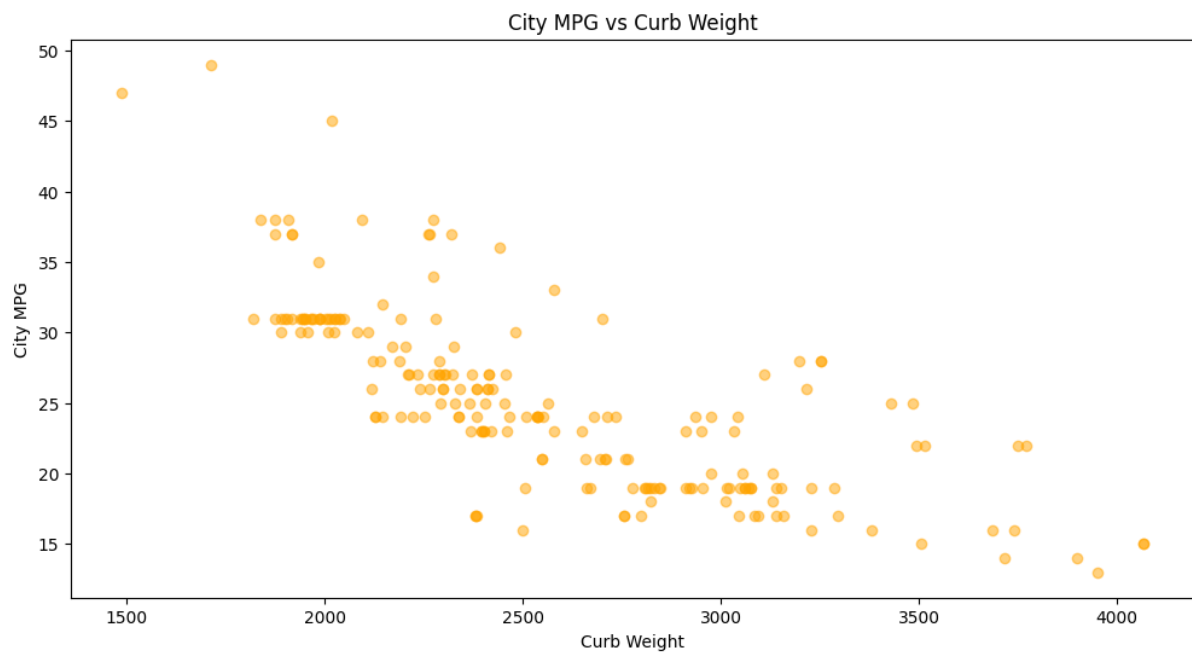
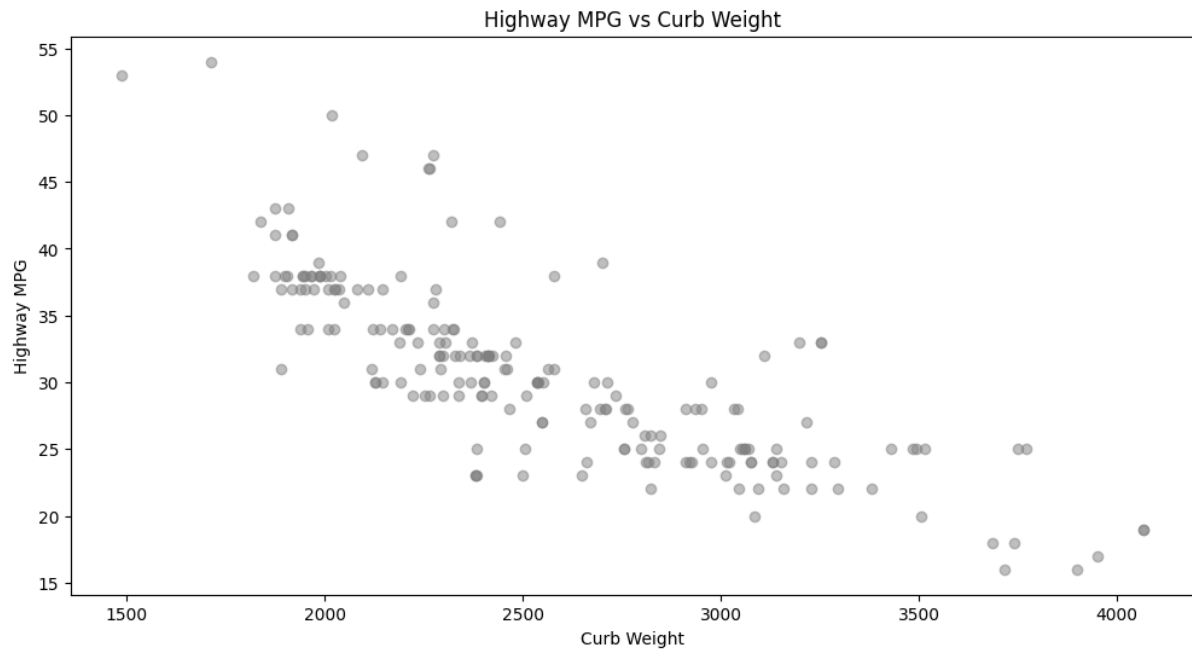


To extend the analysis of fuel efficiency beyond individual vehicles, I calculated the mean city and highway MPG for each manufacturer and visualised the results in a bar chart. This comparison highlighted a clear standout: Chevrolet achieved the highest average MPG in both city and highway driving, outperforming all other manufacturers by a notable margin. At the opposite end of the spectrum, Jaguar recorded the lowest average MPG, reflecting its focus on heavier, performance-oriented models. This manufacturer-level view reinforces earlier findings that fuel economy varies substantially across brands. Notably, there were no significant differences between city and highway MPG within each manufacturer, indicating that brand-level efficiency patterns remain consistent across driving conditions. This was also evident in the previous scatter plots when investigating city and highway MPG vs price.



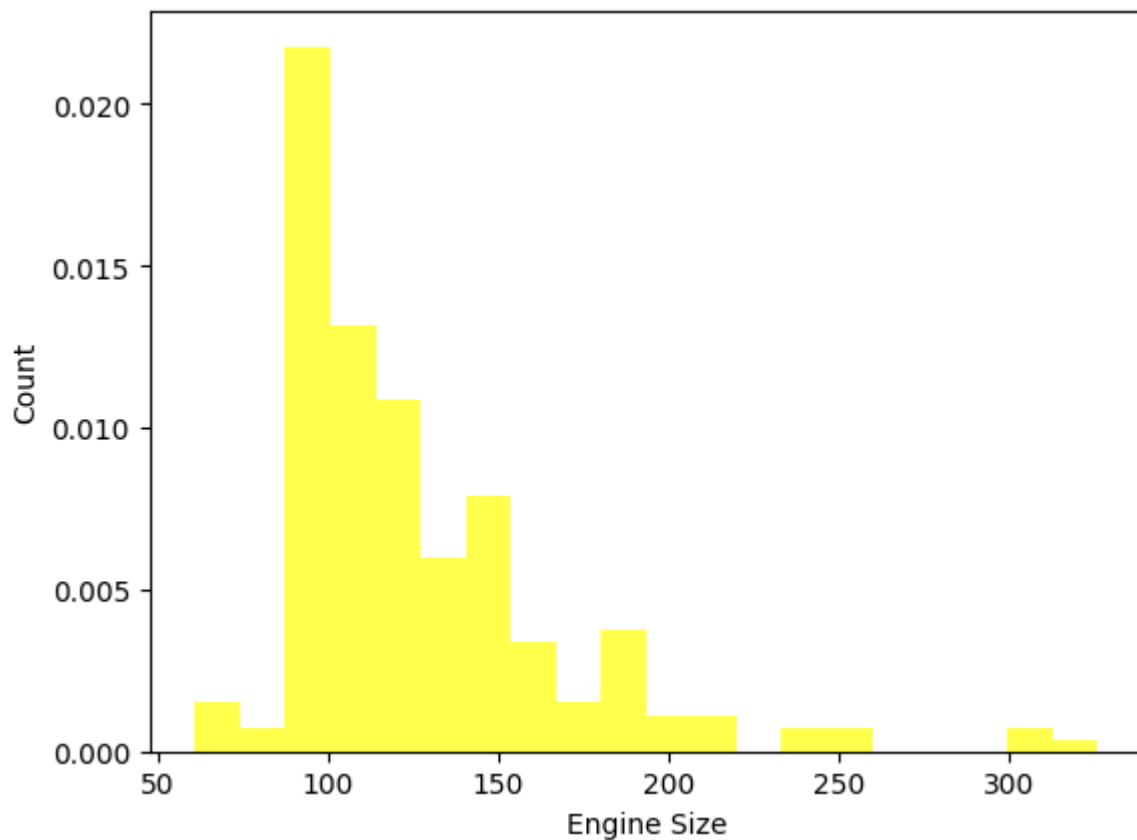


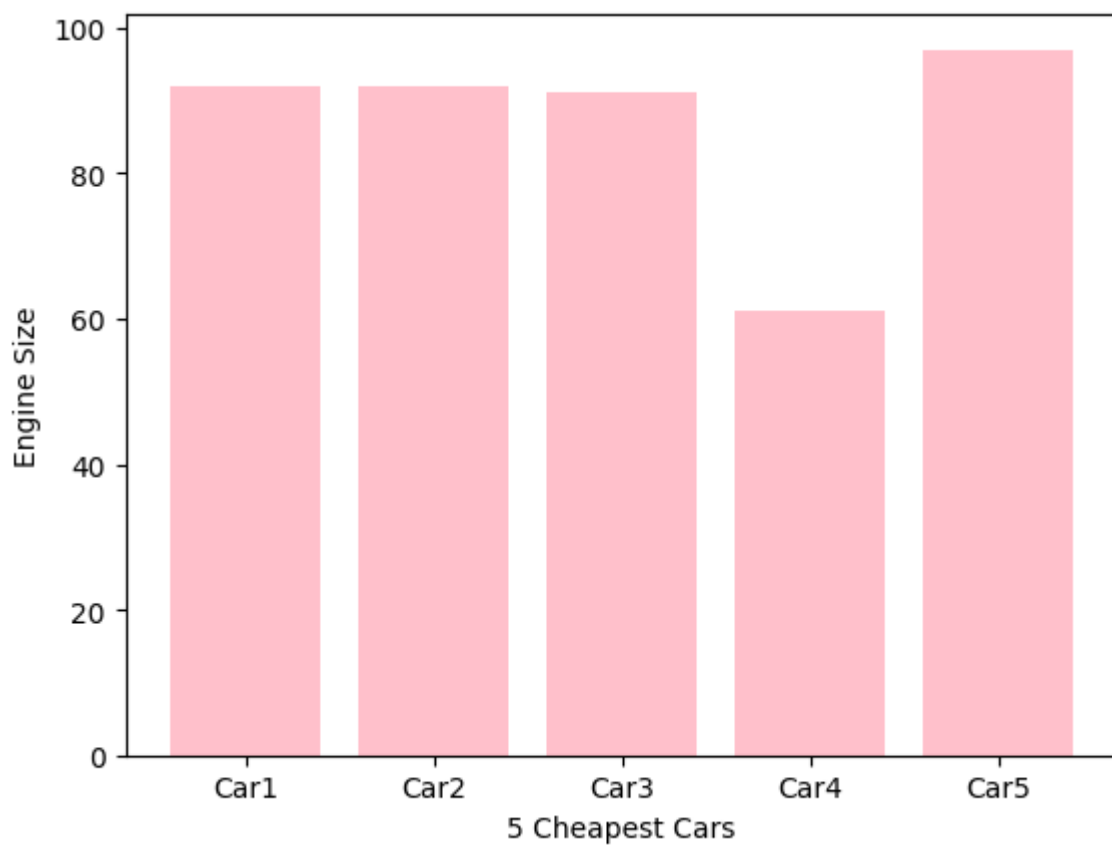
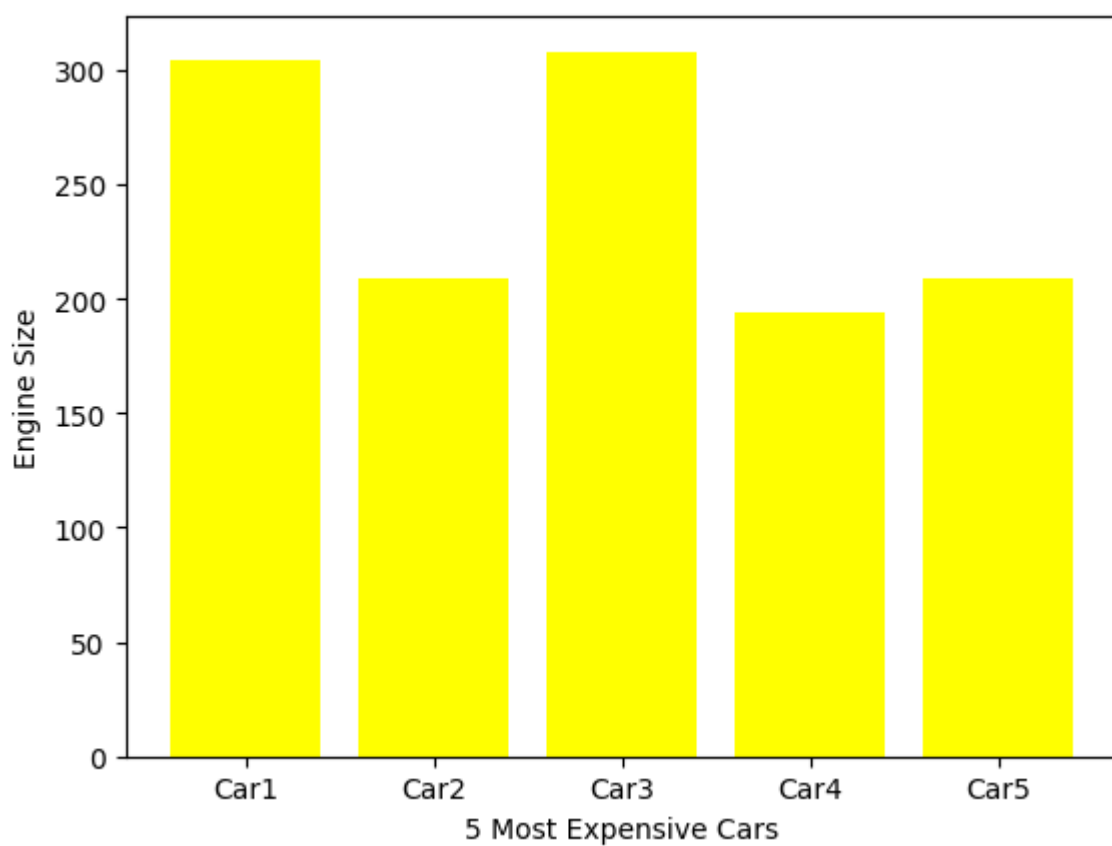
To build on the earlier fuel-efficiency findings, I next examined the relationship between curb weight and city/highway MPG, since vehicle weight is a fundamental determinant of how much energy is required to move the car. Heavier vehicles typically demand more power to accelerate and maintain speed, so exploring this relationship helps clarify whether weight is a key driver behind the MPG patterns already observed. Plotting curb weight against both city and highway MPG confirmed a strong inverse trend: as curb weight increases, fuel efficiency decreases. Lighter cars consistently achieved higher MPG values, while heavier models clustered toward the lower-efficiency end of the scale. This reinforces the idea that structural and design factors play a major role in overall fuel economy.



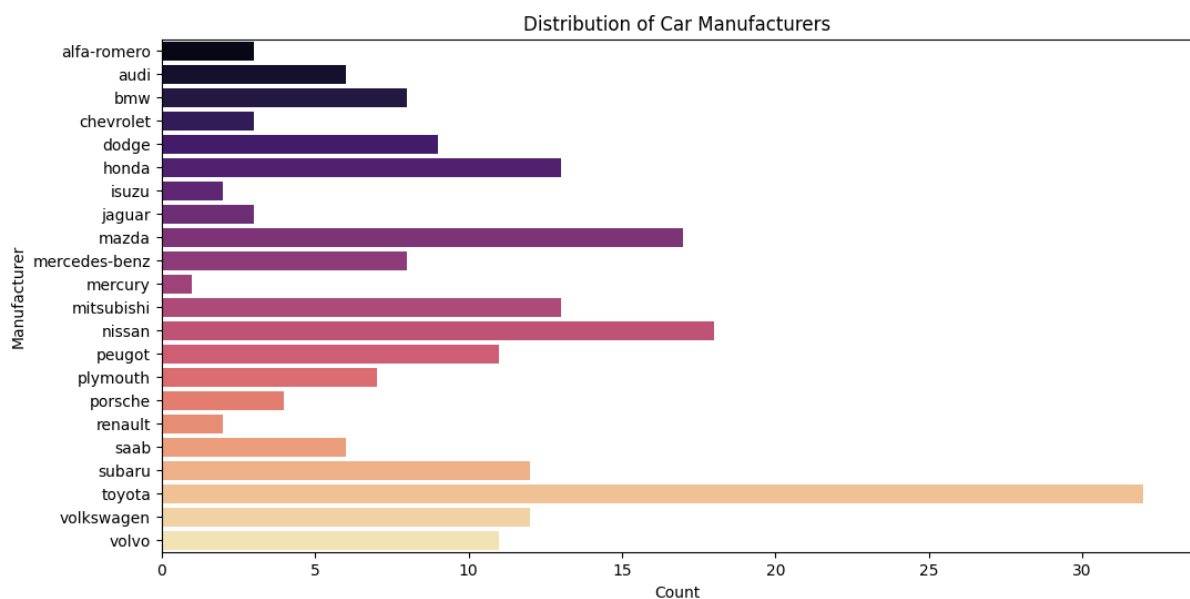
To better understand the performance characteristics represented in the dataset, I next explored the distribution of engine sizes. The histogram revealed that engine sizes in the dataset are heavily concentrated around smaller values, with most vehicles having an engine size close to 100 units. The frequency gradually declines as engine size increases, forming a right-skewed distribution. This pattern suggests that compact and mid-range engines dominate the dataset, while larger engines are relatively uncommon.

I then decided to explore engine sizes in the five most and least expensive cars in the dataset. In the bar chart of the five most expensive cars, engine sizes are substantially larger, ranging roughly from 190 to just over 300 units, and show moderate variation. This suggests that high-priced vehicles tend to feature larger, more powerful engines, consistent with their performance-oriented design. By contrast, the five cheapest cars have much smaller engine sizes, between 60 and 100 units, but with greater variation among them. This indicates that while cheaper cars generally prioritise efficiency and affordability, they differ more widely in engine capacity, possibly reflecting diverse design goals within the low-cost segment. Together, these charts reinforce the broader trend that price correlates positively with engine size, highlighting how performance and power are key contributors to vehicle cost.





I then explored the most popular car manufacturers in the dataset by creating a bar count chart. The bar chart shows that the representation of manufacturers in the dataset is uneven, with some brands appearing far more frequently than others. Subaru has the highest count, making it the most common manufacturer in the dataset, followed by Toyota and Nissan. In contrast, brands such as Isuzu and Mercury appear only once or twice, indicating very limited presence. Overall, the chart reveals that the dataset is dominated by a handful of mainstream manufacturers, while several luxury brands are included only sparsely.



Conclusion

Overall, the analysis makes it clear that vehicle price and fuel efficiency are closely linked, with higher-priced cars consistently showing lower city and highway MPG. This relationship aligns with the broader patterns observed throughout the dataset: expensive vehicles tend to feature larger engines, higher horsepower, and greater curb weight, all of which naturally reduce fuel economy. In contrast, the cheapest cars generally have smaller, lighter, and less powerful engines, resulting in much higher MPG values. The influence of factors such as curb weight and engine size highlights that fuel efficiency is shaped by a combination of design and performance characteristics, not price alone. It's also important to note that the dataset itself is dominated by lower-priced vehicles with smaller engines, which may have subtly influenced the overall trends by making efficiency-focused models more prominent in the analysis.

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